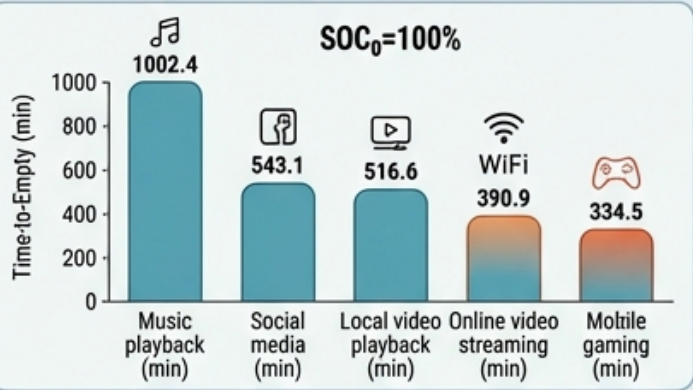


A Continuous-Time SOC Model with Thermal Feedback for Smartphone Battery Analysis

We model battery SOC as a coupled power–thermal dynamical system to predict **Time-to-Empty (TTE)** and identify dominant drain drivers across common usage scenarios.

Main Results + Insights



Insights: "who kills the battery" on the scenario

- **[Scenario dominance]** The primary drain driver shifts across scenarios, so "one-size-fits" savings are misleading.
- **[Gaming]** CPU is the dominant driver (51.5%), amplified by thermal stress, sustained high load shortens TTE the most.
- **[Streaming]** Online streaming is a **Screen 49.4% + CPU 38.9%** combo—it's not "just network".
- **[Social & Local video]** Screen dominates: **Social screen 56.3%**, **Local video screen 76.0%** → brightness control 56.3% = lowest radiant power has outsized impact here.
- **[Music]** Surprisingly, **Network 54.9%** due to network tail energy, though the world for outsized impact here.
- **[SOC scaling]** Initial SOC mainly scales total time, it does not change which analysis which scenario is best/worst.

Practical Recommendations (Actionable + Impact)

Everyday User Actions

- Do:** Lower screen brightness (or enable adaptive brightness) for social browsing. **↑ HIGH**
Why: Screen is the largest share in Social (56.3%) and Local video (76.0%).
- Do:** Avoid long, continuous mobile gaming sessions, take cooldown breaks. **↑ HIGH**
Why: Gaming is CPU dominant (51.5%) thermal stress, shortening TTE most.
- Do:** Prefer WiFi when high-speed cellular isn't needed, reduce background syncing. **↑ MED**
Why: Network tail energy can dominate in high scenarios (Music: Network 54.9%).
- Do:** When battery is low, avoid high-power tasks (gaming/streaming), switch, switch to low intensity use. **↑ MED-HIGH**
Why: Under low SOC, "high power per minutes" has the worst cost benefit.

System / OS Policy Ideas

- Policy:** Thermal-aware performance scaling (smooth throttling) + aggressive brightness governance in screen-dominant apps.
- Trigger/Impact:** High temps/load → MED-HIGH impact

Network / Thermal Strategy

- Strategy:** Tail-energy-aware networking (batch downloads, avoid frequent small transfers; prefer WiFi).
- Trade-off/Impact:** Slight latency vs lower power → MED impact

Model: Continuous-time coupled SOC–power–thermal framework with network tail energy and thermal throttling feedback.